Aim

To connect four Cisco routers and configure static routing between different networks using Cisco Packet Tracer.

Description

4-Router Configuration

Four Cisco 2811 routers are connected in a ring topology using serial cables. Each router is connected to a switch and PCs. IP addresses and default gateways are assigned to all devices, and static routes are configured on the routers to enable communication between all four networks. Connectivity is verified using ICMP PDU and the ping command**.**

between the two LANs is established successfully.

Network Configuration

Four routers are connected in a **ring topology**. Each router connects one LAN containing two PCs.

Device	Interface	IP Address	Default Gateway
PC5	FastEthernet	192.168.5.2	192.168.5.1
PC6	FastEthernet	192.168.5.3	192.168.5.1
Router2	LAN	192.168.5.1	—
Router2	Serial 0/1/0	10.1.1.1	—
Router2	Serial 0/1/1	10.1.4.2	—
Router3	Serial 0/1/0	10.1.1.2	—
Router3	Serial 0/1/1	10.1.2.1	—
Router3	LAN	192.168.6.1	—
PC7	FastEthernet	192.168.6.2	192.168.6.1
PC8	FastEthernet	192.168.6.3	192.168.6.1
Router5	Serial 0/1/1	10.1.2.2	—
Router5	Serial 0/1/0	10.1.3.1	—
Router5	LAN	192.168.8.1	—

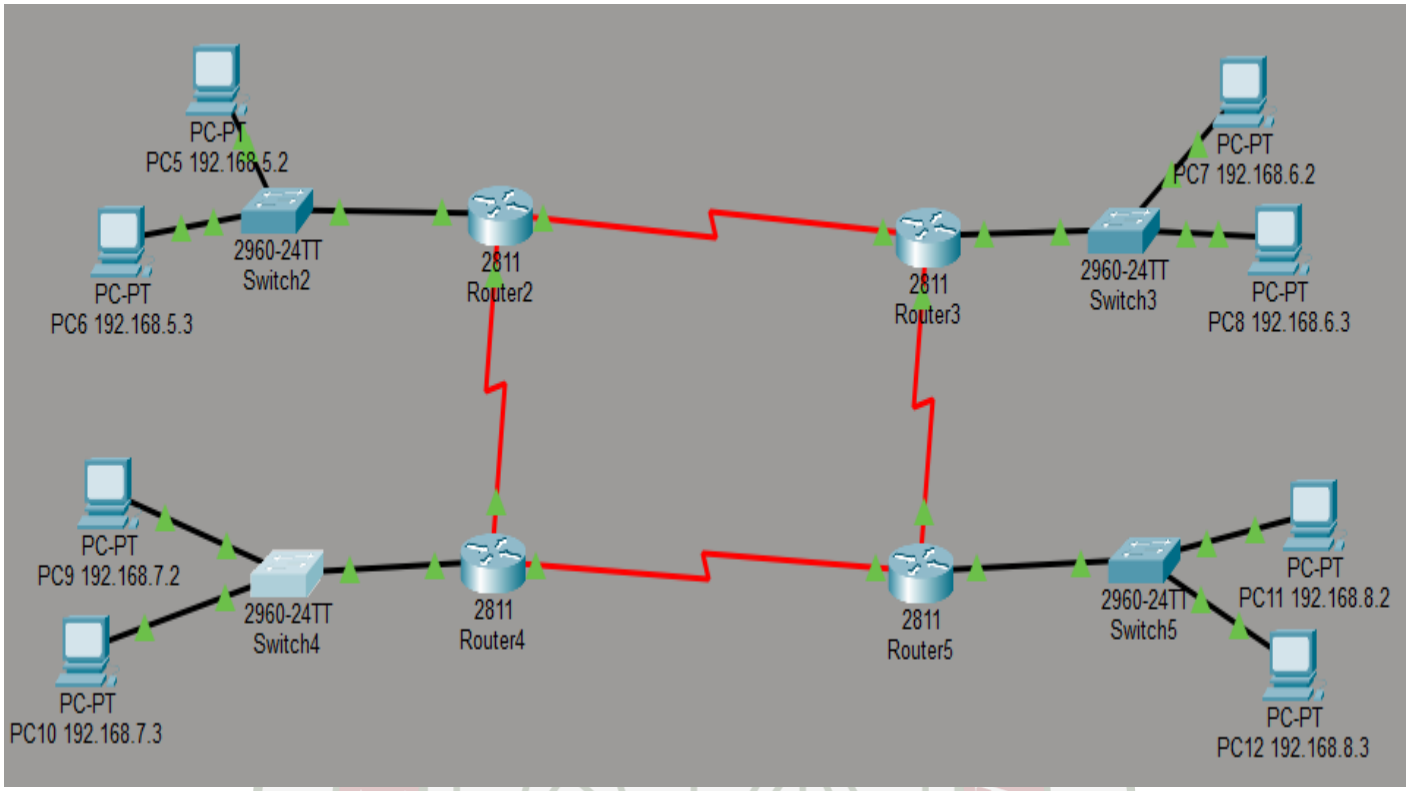
PC11	FastEthernet	192.168.8.2	192.168.8.1
PC12	FastEthernet	192.168.8.3	192.168.8.1
Router4	Serial 0/1/0	10.1.3.2	—
Router4	Serial 0/1/1	10.1.4.1	—
Router4	LAN	192.168.7.1	—
PC9	FastEthernet	192.168.7.2	192.168.7.1
PC10	FastEthernet	192.168.7.3	192.168.7.1

Router Serial Connection Table

Connection	**Interface 1**	**IP**	**Interface 2**	**IP**
Router2 ↔ Router3	R2 Serial 0/1/0	10.1.1.1	R3 Serial 0/1/0	10.1.1.2
Router3 ↔ Router5	R3 Serial 0/1/1	10.1.2.1	R5 Serial 0/1/1	10.1.2.2
Router5 ↔ Router4	R5 Serial 0/1/0	10.1.3.1	R4 Serial 0/1/0	10.1.3.2
Router4 ↔ Router2	R4 Serial 0/1/1	10.1.4.1	R2 Serial 0/1/1	10.1.4.2

Subnet Mask: 255.255.255.0

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Static Routing

Static routes were configured on all four routers using the ip route command so that each router could reach the remote LANs through the appropriate next-hop router

Router2

```
Router(config)#ip route 192.168.7.0 255.255.255.0 10.1.4.1
Router(config)#ip route 192.168.6.0 255.255.255.0 10.1.1.2
Router(config)#ip route 192.168.8.0 255.255.255.0 10.1.1.2
Router(config)#
```

Router3

```
Router(config)#ip route 192.168.5.0 255.255.255.0 10.1.1.1
Router(config)#ip route 192.168.8.0 255.255.255.0 10.1.2.2
Router(config)#ip route 192.168.7.0 255.255.255.0 10.1.2.2
Router(config)#
```

Router4

```
Router(config)#ip route 192.168.5.0 255.255.255.0 10.1.4.2
Router(config)#ip route 192.168.8.0 255.255.255.0 10.1.3.1
Router(config)#ip route 192.168.6.0 255.255.255.0 10.1.3.1
Router(config)#
```

Router5

```
Router(config)#ip route 192.168.7.0 255.255.255.0 10.1.3.2
Router(config)#ip route 192.168.6.0 255.255.255.0 10.1.2.1
Router(config)#ip route 192.168.5.0 255.255.255.0 10.1.2.1
Router(config)#
```

Output

ICMP PDU testing was performed between different LANs.

PDU List Window

Fire	Last Status	Source	Destination	Type
	Successful	PC5 192.168.5.2	PC7 192.168.6.2	ICMP
	Successful	PC6 192.168.5.3	PC10 192.168.7.3	ICMP
	Successful	PC6 192.168.5.3	PC12 192.168.8.3	ICMP

Justification

All tested ICMP packets were successfully delivered between different LANs. This confirms that the four routers have the required static routes and that packets can traverse multiple routers to reach remote networks. The successful PDU results verify that the **four-router static routing topology is functioning correctly**.

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